SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence Machine Learning

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	D. Sri Anjaneya	**Reg. No.:**	192425219
Section / Batch:	AI & ML / 2024	**Date:**	12/08/2026

Code of Conduct

I _D. Sri Anjaneya / 192425219_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___D. Sri Anjaneya___

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Annexure A – Industry Problem Solving Task

1. A company is developing an AI-powered customer support chatbot for banking services. The chatbot uses Context-Free Grammar (CFG) for parsing user queries. However, it faces issues such as:
Incorrect interpretation of ambiguous sentences
Failure to handle subject–verb agreement
Inefficient parsing for long queries Example
user query:
"Show me the transactions with the card from last month"
a) Analyze the ambiguity present in the given sentence using CFG
b) Evaluate the limitations of CFG in handling real-world conversational queries
c) Evaluate and propose an improved solution using PCFG, feature structures, or Earley parsing
d) Justify how your proposed method improves accuracy and efficiency in an industry setting

2. A voice assistant system (like Alexa/Siri-type applications) processes spoken commands using parsing techniques. The system currently uses top-down parsing, but faces:
Backtracking issues
Poor handling of incomplete or ambiguous input
Difficulty in real-time response Example
command:
"Book a flight to Delhi with a window seat"
a) Analyze how the given sentence can lead to multiple parse structures
b) Analyze the limitations of top-down parsing in this real-time application
c) Analyze how Earley parsing can improve handling of ambiguity and partial inputs
d) Compare and analyze the performance of top-down vs Earley parsing for such industry use cases

3. A healthcare company is developing an AI system to analyze patient medical reports and automatically extract critical information such as diagnosis, prescribed treatments, and follow-up actions.
The system must handle:
Complex and lengthy medical sentences
Ambiguity in sentence structure
Accurate subject-verb agreement
Relationships between medical actions and entities (sub-categorization)
Real-time processing for hospital use Example input:
"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."
The current system, based only on basic Context-Free Grammar (CFG), fails to correctly interpret relationships and produces inconsistent outputs.
a) Design a complete NLP architecture integrating:
CFG for syntactic structure
Probabilistic CFG (PCFG) for ambiguity resolution
Feature Structures for agreement handling
Efficient parsing techniques
b) Develop a step-by-step workflow showing how the system processes the given medical sentence from input to structured output (diagnosis, action, location). c) Create a method to:
Incorporate sub-categorization frames for medical verbs
Handle real-time data processing
Improve accuracy and scalability in a hospital environment

1. Banking Customer Support Chatbot Using CFG

a) Ambiguity in the Given Sentence Using CFG

The sentence is:

"Show me the transactions with the card from last month."

A CFG can represent the sentence using rules such as:

- S → VP
- V p → V NP
- NP → Det N PP
- NP → Det N

The sentence contains **prepositional phrase (PP) attachment ambiguity**.

The phrase **"with the card"** can be interpreted in two ways:

Interpretation 1 – Card used for transactions:

Show me [the transactions [with the card]] [from last month].

Here, **"with the card"** modifies **transactions** and indicates that the transactions were made using a particular card.

Interpretation 2 – User wants transactions together with the card:

Show me [the transactions] [with the card] [from last month].

Here, the phrase may be interpreted as an additional object or context associated with the request.

Similarly, **"from last month"** must be associated with the correct noun or event. In a banking application, the intended meaning is generally:

Transactions → performed using the card → during last month

A basic CFG mainly describes whether a sentence follows grammatical rules. It does not inherently understand the user's actual banking intention.

Subject–Verb Agreement Problem

CFG rules can generate grammatically valid structures, but basic CFG does not automatically enforce features such as:

- Singular/plural agreement.
- Person agreement.
- Tense.
- Number agreement.

For example:

"Show me the transactions…"

is grammatically correct, but a basic grammar could also generate structurally valid combinations without checking all linguistic features.

b) Limitations of CFG in Real-World Conversational Queries

Although CFG is useful for defining syntactic structures, it has several limitations in banking chatbots.

1. Ambiguity

CFG may generate multiple valid parse trees for an ambiguous sentence but cannot reliably determine which interpretation is intended.

2. Lack of Context

A basic CFG does not understand conversation history.

For example:

User: “Show my transactions.”

User: “Only the card transactions.”

The second sentence depends on the previous context, which basic CFG cannot naturally represent.

3. Subject–Verb Agreement

Basic CFG rules do not inherently encode grammatical features such as number and person.

4. Long Sentences

Long conversational queries can produce a large number of possible parse trees, increasing parsing complexity.

5. Informal Language

Bank customers may use:

“Show last month card spends”

instead of a formally structured sentence. A strict CFG may fail to parse such input.

6. Domain Understanding

CFG understands syntactic structure but does not inherently understand concepts such as:

transaction → banking operation

card → payment instrument

last month → temporal constraint

Therefore, additional semantic processing is required.

c) Improved Solution Using PCFG, Feature Structures and Earley Parsing

A practical solution should combine several techniques instead of depending only on CFG.

Proposed Architecture

User Query → Tokenization → Feature Analysis → Earley Parsing → PCFG Ranking → Semantic Interpretation → Banking API → Response

1. PCFG

A **Probabilistic Context-Free Grammar** assigns probabilities to grammar rules.

If multiple parse trees are possible, the system can select the most probable interpretation based on training data.

For example:

The higher-probability interpretation can be selected.

2. Feature Structures

Feature structures can represent grammatical information such as:

- Number
- Person
- Tense
- Agreement
- Semantic attributes

For example:

and

can be matched to ensure correct agreement.

3. Earley Parsing

Earley parsing is suitable for natural-language grammars because it can handle:

- Ambiguous grammars.
- Left recursion.
- Partial input.
- Complex sentence structures.

It uses dynamic programming to avoid repeatedly parsing the same portions of a sentence.

Recommended Combination

CFG → defines grammatical structure

Feature Structures → enforce agreement

PCFG → selects likely interpretation

Earley Parser → efficiently handles complex/ambiguous input

d) Industry Benefits

The proposed method improves the banking chatbot in several ways.

1. **Better ambiguity resolution:** PCFG ranks competing interpretations.
2. **Improved grammatical accuracy:** Feature structures handle agreement.
3. **Efficient parsing:** Earley parsing avoids unnecessary repeated work.
4. **Better conversational support:** Partial and incomplete queries can be handled more effectively.
5. **Banking-domain integration:** Semantic analysis can map parsed queries to banking operations.
6. **Scalability:** The architecture can support large numbers of customer queries.

For the example query, the final structured representation could be:

INTENT: SHOW_TRANSACTIONS

OBJECT: CARD_TRANSACTIONS

TIME: LAST_MONTH

Conclusion

A basic CFG provides the syntactic foundation, but an industry chatbot requires more than grammar. Combining **PCFG + Feature Structures + Earley Parsing + Semantic Analysis** provides better ambiguity resolution, grammatical accuracy and efficient processing, making the system more suitable for real banking environments.

2. Voice Assistant Using Top-Down and Earley Parsing

a) Multiple Parse Structures

The command is:

“Book a flight to Delhi with a window seat.”

The sentence contains a possible **prepositional phrase attachment ambiguity**.

Interpretation 1

“with a window seat” describes the flight booking preference.

Book

└── Flight

├── to Delhi

└── with a window seat

Meaning:

Book a flight to Delhi and request a window seat.

This is the most likely interpretation.

Interpretation 2

The phrase **“with a window seat”** could potentially be attached differently in the syntactic structure, creating another parse tree.

The problem becomes more important in voice systems because users often speak naturally, omit words or pause unexpectedly.

Semantic Representation

The intended meaning can be represented as:

ACTION: BOOK

OBJECT: FLIGHT

DESTINATION: DELHI

SEAT_PREFERENCE: WINDOW

b) Limitations of Top-Down Parsing

The current voice assistant uses **top-down parsing**, where parsing starts from the start symbol and attempts to generate a structure matching the input.

1. Backtracking

If the parser chooses an incorrect production rule, it may need to backtrack and try another possibility. This increases processing time.

2. Ambiguous Input

A sentence with multiple possible structures can result in several parsing paths.

3. Incomplete Input

Voice assistants frequently receive incomplete commands such as:

“Book a flight to…”

Top-down parsing may struggle because the sentence has not yet been completed.

4. Real-Time Requirement

Voice assistants need fast responses. Excessive backtracking can increase latency.

5. Conversational Speech

Users may pause, restart or correct themselves:

“Book a flight to Delhi… actually Mumbai.”

A rigid top-down approach may have difficulty handling such revisions.

6. Repeated Computation

The parser may repeatedly process the same sentence fragments while exploring different alternatives.

c) How Earley Parsing Improves the System

Earley parsing is a **dynamic-programming parsing technique** that is particularly useful for natural-language grammars.

It maintains partial parsing states and avoids unnecessarily repeating work.

Three Main Operations

1. Prediction

Predicts grammar rules that may be needed.

2. Scanning

Matches predicted terminals with the input words.

3. Completion

Completes a grammar rule when all required components have been recognized.

Advantages for Voice Assistants

- Handles ambiguous grammars.
- Supports left-recursive grammar rules.
- Processes partial input.
- Avoids unnecessary repeated parsing.
- Maintains multiple possible interpretations.
- Can work effectively with natural-language grammars.

For:

“Book a flight to Delhi with a window seat”

Earley parsing can maintain possible structures and allow a probabilistic or semantic layer to select the most appropriate interpretation.

d) Top-Down vs Earley Parsing

Feature	Top-Down Parsing	Earley Parsing
Parsing direction	Starts from start symbol	Processes input while maintaining states
Ambiguity	Can require extensive backtracking	Handles ambiguity more systematically
Backtracking	Often required	Reduced through dynamic programming
Partial input	Less suitable	Better suited
Left recursion	Problematic in common top-down implementations	Supported
Long sentences	Can become inefficient	More suitable for complex grammar
Real-time use	Can suffer from latency	Better suited with optimization
Natural-language applications	Limited for highly ambiguous input	Stronger for general CFG-based NLP

Industry Solution

A practical voice assistant can use:

Speech Recognition → Tokenization → Earley Parser → PCFG/Intent Ranking → Semantic Interpretation → Action Execution

For the given command:

Intent: BOOK_FLIGHT

Destination: DELHI

Seat: WINDOW

3. Healthcare NLP System for Medical Report Analysis

a) Complete NLP Architecture

The healthcare system should combine **CFG, PCFG, Feature Structures, efficient parsing and Semantic analysis**.

Proposed Architecture

Medical Report

↓

Text Preprocessing

↓

Tokenization & POS Tagging

↓

Feature Extraction

↓

CFG Parsing

↓

Feature Structures

↓

Earley / Efficient Parsing

↓

PCFG Ambiguity Resolution

↓

Sub-categorization & Semantic Role Analysis

↓

Named Entity Recognition

↓

Structured Medical Information

↓

Hospital Database / Decision Support

b) Step-by-Step Processing of the Given Sentence

Input:

"The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

Step 1 – Tokenization

The sentence is divided into tokens:

The | doctor | who | reviewed | the | patient | last | week | recommends | starting | medication | and | scheduling | a | follow-up | visit | in | Chennai

Step 2 – Syntactic Analysis

The parser identifies:

- Main subject → **doctor**
- Relative clause → **who reviewed the patient last week**
- Main verb → **recommends**
- Action 1 → **starting medication**
- Action 2 → **scheduling a follow-up visit**
- Location → **Chennai**

Step 3 – Feature Structure

The system identifies:

Subject:

Entity = Doctor

Number = Singular

Person = Third

Verb:

recommends

Number = Singular

Tense = Present

Thus, **doctor → recommends** satisfies subject–verb agreement.

Step 4 – Relative Clause Analysis

The phrase:

"who reviewed the patient last week"

modifies **doctor**.

Therefore:

Doctor

└── reviewed → Patient

└── Time → Last Week

The system should not incorrectly attach **last week** to the recommendation.

Step 5 – Semantic Role Assignment

Entity	**Role**
Doctor	Agent

Entity	Role
Patient	Patient/Experiencer
Medication	Treatment/Theme
Follow-up Visit	Medical Event
Chennai	Location
Last Week	Time

c) Method for Sub-Categorization, Real-Time Processing and Scalability

1. Sub-Categorization Frames

Create a domain-specific medical verb lexicon.

Examples:

prescribe(Doctor, Medicine, Patient)

recommend(Doctor, Treatment/Action)

diagnose(Doctor, Disease, Patient)

schedule(Staff, MedicalEvent, Time/Location)

monitor(Doctor/Nurse, Patient)

When a sentence is parsed, the system checks whether the detected arguments match the expected verb frame.

This helps prevent incorrect relationships.

2. Real-Time Processing

For hospital environments, the system should use an efficient pipeline:

Input → Fast Preprocessing → Incremental Parsing → Semantic Extraction → Structured Output

Techniques include:

- Incremental parsing.
- Efficient Earley parsing.
- Cached grammar rules.
- Parallel processing.
- Batch processing where appropriate.
- GPU acceleration for neural components.

The system should also prioritize urgent clinical information.

3. Scalability

For large hospitals, the architecture can use:

Hospital Records

↓

Data Processing Layer

↓

NLP Services

↓

Parsing + Semantic Analysis

↓

Structured Medical Data

↓

Hospital Database

The NLP services can be deployed as independent scalable services so that multiple medical reports can be processed simultaneously.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	Understanding of Industry Problem	Application of Engineering Concepts	Solution Design	Use of Tools	Analysis	Professionalism	Sub. Total	Total %
1								
2								
3								

Faculty In-charge	Course Coordinator	Program Director

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____